Hashanpreet Singh Dhaliwal

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EDUCATION

B.Eng. Aerospace Engineering (Aerodynamics, Propulsion & Vehicle Performance) **Carleton University,** Ottawa, ON

GPA: 8.56/12 (B) / Dean's Honor List / General In-course Scholarship

PROFESSIONAL EXPERIENCE

Collins Aerospace (RTX), Ottawa, ON | Coop Student **Sept 2023 – August 2024**

- **System Verification & Compliance:** Inspected, tested, and debugged printed circuit boards (PCBs) for radio and modem systems within ESD-controlled lab environments.
- **Manufacturing Integration:** Supported the assembly of complex sub-assemblies and final products, contributing to the integration of specialized radio hardware.
- **Testing & Analysis:** Assisted in emissions testing at TÜV SÜD; performed detailed result analysis and presented technical findings to multidisciplinary engineering teams.
- **CAD Documentation:** Prepared detailed CAD drawings and 3D printed models using **SolidWorks** and **CATIA** to support ongoing hardware development projects.

TECHNICAL PROJECTS

Gas Turbine Performance Optimization (Jet-Cat P500) (Capstone Project)

- **Computational Analysis:** Developed multiple **MATLAB** programs to execute diverse thermodynamic approaches for solving engine parameters and establishing correlations between component efficiencies.
- **Comparative Engineering:** Analyzed performance variations and efficiency trade-offs across different propulsion architectures, including **turbojet** and **turbofan** configurations.
- **Reverse Engineering:** Modeled the Jet-Cat P500 turbojet using available specifications and optimized component efficiencies (compressor, combustor, turbine) by varying Turbine Inlet Temperature (TIT).
- **Parametric Studies:** Conducted detailed studies varying compressor pressure ratios (CPR) and TIT to minimize **TSFC** and maximize specific thrust.

V2500 Turbofan Preliminary Design (Aerospace Propulsion)

- **Cycle Modeling:** Utilized **MDIDS-GT** to simulate engine performance at take-off, generating stage-specific temperature and pressure data.
- **Component Design:** Designed compressor and turbine mean-line stages and developed blade airfoils/disks to meet preliminary stress analysis criteria.
- **Aerodynamic Validation:** Performed CFD analysis of the fan stage to validate design accuracy against group stage pressure ratios.

Retractable Landing Gear System & Flap Systems Design (Systems & Design)

- **CAD & 3D Modeling:** Developed detailed 3D models and assembly drawings for Nose Landing Gear (NLG) and Main Landing Gear (MLG), illustrating precise kinematics for both deployed and retracted configurations.
- **Aerodynamic & Performance Analysis:** Executed comprehensive calculations for landing gear and flap drag, strut sizing, and vertical tire travel. Determined optimal tire sizes and required runway lengths based on aircraft performance parameters.
- **Safety & Systems Engineering:** Conducted Failure Modes and Effects Analysis (FMEA) to identify critical risks and ensure reliable system deployment under emergency conditions.
- **Constraint-Driven Design:** Defined and managed system and design constraints, performing a weighted trade study to compare hydraulic vs. electrical actuation systems based on performance, reliability, and safety metrics.

Computer-Aided Design and Manufacturing (CAD/CAM) Project, **Toolpath Generation:** Utilized SolidWorks CAM to generate 2.5D and 3-axis milling toolpaths for aerospace components.

- **G-Code Optimization:** Translated CAD models into machine-readable **G-code and M-code**, optimizing feed rates and spindle speeds for different materials (Aluminum 6061/Stainless Steel).
- **Simulation & Verification:** Performed virtual machining simulations to detect and prevent tool collisions, ensuring the accuracy of the final part geometry before physical production.
- **Fixturing Design:** Designed custom work-holding fixtures in **SolidWorks** to ensure part stability and repeatability during multi-op machining processes.

PUBLICATION

Treatment of Distillery Spent Wash for Irrigation Purposes Journal of Advance Research in Applied Science https://nnpub.org/index.php/AS/article/download/644/579/

- **Impact:** Investigated sand as an adsorbent for effluent treatment, recording chemical data (pH, COD, TDS) to improve sustainable agricultural irrigation.

BUSINESS OWNER | Customized Phone Cases Printing Machine

- **Operations & Deployment**: Managed the end-to-end commissioning and strategic placement of an automated "Phone cases" unit within a high-traffic shopping center.
- **Process Optimization:** Optimized on-demand manufacturing workflows to maintain a 3-minute production cycle for over 100 unique device configurations.
- **Stakeholder & Resource Management**: Oversaw vendor relations, supply chain logistics, and visual marketing strategies to maximize ROI and ensure 99% machine uptime.

SKILLS

- **Analysis:** ANSYS Fluent (CFD), MATLAB, MDIDS-GT, FEA simulations.
- **CAD/Design:** SolidWorks, CATIA, Fusion 360, Autodesk Inventor.
- **Programming:** Python (Automation & Visualization), C++, MS Excel, CNC Programming (G-codes).
- **Data Science/ML:** Large Language Models (LLMs), Prompt Engineering, Neural Networks (CNNs).
- **Communication:** Professional fluency in English; intermediate proficiency in French.